

Answer Grading for Short Answers

Dataset: SemEval-2013 Community Question Answering (Task 3)

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1. Problem Statement

Automatic answer grading is the task of assessing the quality of a short answer given in response to a question. This is a core problem in Natural Language Processing (NLP) with applications in intelligent tutoring systems, automated assessment platforms, and community question-answering (CQA) services.

In this project, answer grading is framed as a three-class text classification problem. Given an answer text, the model must predict one of three quality labels: Bad (-1), Neutral (0), or Good (1). This multi-class formulation more accurately reflects real-world answer quality, which exists on a spectrum rather than a binary good/bad distinction.

2. Dataset

We use the SemEval-2013 Community Question Answering dataset (Task 3), part of the SemEval shared task series on semantic evaluation. The dataset contains answers collected from online community forums, annotated by human judges with three quality labels: -1 (Bad), 0 (Neutral/Potentially Useful), and 1 (Good).

Split	Bad (-1)	Neutral (0)	Good (1)	Total
Train	1444	4550	3622	8316
Development	340	737	573	1650
Test	559	1513	1473	3545

Table 1: Class distribution per split.

Each instance contains a text field (the raw answer string) and a label field (-1, 0, or 1). The text is user-generated and contains noise such as URLs, @mentions, hashtags, and informal language, requiring preprocessing before model training.

3. Methodology

3.1 Preprocessing

Each answer text was processed through the following pipeline:

- Lowercasing and removal of URLs, @mentions, and #hashtags
- Stripping punctuation and non-alphabetic characters
- Stopword removal using the NLTK English stopwords list
- Lemmatization using the WordNet Lemmatizer
- Removal of empty strings produced by cleaning

3.2 Feature Extraction

Cleaned text was converted to numerical features using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization. We extracted the top 10,000 features using unigrams and bigrams (`ngram_range = (1, 2)`), with sublinear TF scaling and a minimum document frequency of 2. The vectorizer was fitted exclusively on the training set and applied to development and test sets to prevent data leakage.

3.3 Models

We trained and compared two linear classifiers well-suited to high-dimensional sparse TF-IDF feature spaces:

- Logistic Regression — a probabilistic multi-class classifier using the one-vs-rest (OvR) strategy with L2 regularization ($C = 1.0$), trained for a maximum of 1,000 iterations.
- LinearSVC — a Support Vector Classifier using a linear kernel ($C = 1.0$) with one-vs-rest multi-class handling, trained for a maximum of 2,000 iterations.

Both models natively support multi-class classification and serve as strong baselines. The best model was selected based on weighted F1 score on the test set, which accounts for class imbalance across the three labels.

4. Results

Table 2 reports performance of both models on the development and test sets. Weighted F1 is the primary metric as it accounts for the unequal distribution of Bad, Neutral, and Good answers.

Model / Split	Accuracy	Precision	Recall	F1 (weighted)
Logistic Regression — Dev	62.72%	63.72%	62.72%	0.601608
Logistic Regression — Test	64.93%	67.66%	64.93%	0.629668

Model / Split	Accuracy	Precision	Recall	F1 (weighted)
LinearSVC — Dev	62.60%	62.42%	62.60%	0.615716
LinearSVC — Test	64.09%	64.61%	64.09%	0.634110

Table 2: Model performance across splits.

The 3×3 confusion matrix for the best-performing model (LinearSVC) on the test set is shown in Figure 1. The diagonal cells represent correct predictions for each class. The matrix shows the model performs strongest on Neutral (0) answers, which dominate the training data, and struggles most with Bad (-1) answers, many of which are misclassified as Neutral.

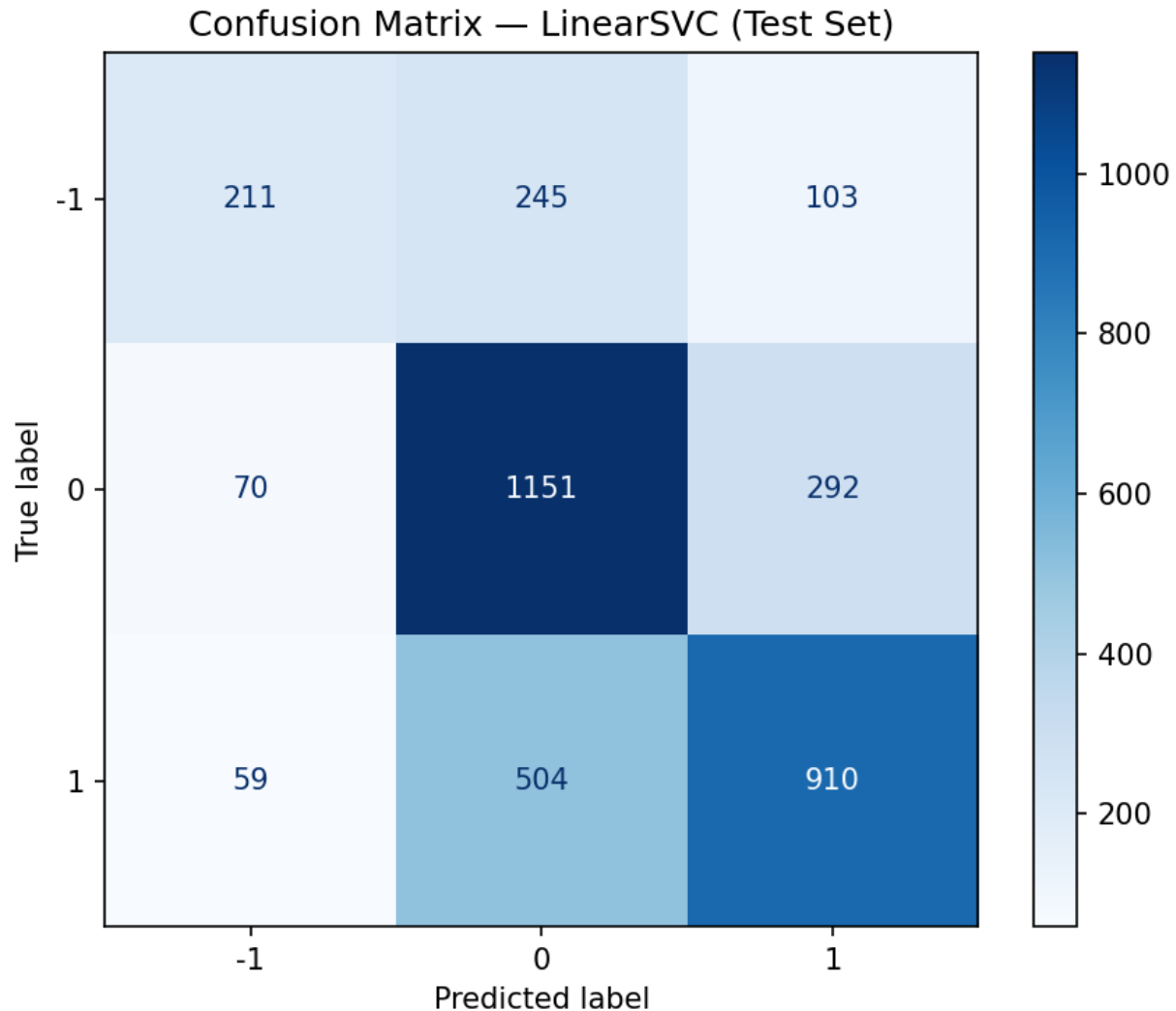


Figure 1: Confusion matrix for LinearSVC on the test set (classes: -1 = Bad, 0 = Neutral, 1 = Good).

4.1 Per-Class Performance

The classification report reveals notable variation in performance across the three classes. The Neutral class (0) achieves the highest F1 due to its larger representation in the training data. The Good class (1) is moderately well-detected, while the Bad class (-1) shows the lowest recall, indicating

that bad answers are frequently confused with neutral ones — a common challenge in ordinal multi-class tasks where adjacent classes share overlapping vocabulary.

4.2 Error Analysis

Inspection of misclassified examples (saved to results/errors.csv) revealed three recurring patterns. First, very short answers (one or two words) provided insufficient signal for the TF-IDF representation to determine quality. Second, bad answers written in a formally correct style were often misclassified as neutral or good, as surface-level features did not capture relevance to the question. Third, good answers using rare or technical vocabulary were underweighted, since TF-IDF penalises terms appearing in few training documents.

5. Discussion

The TF-IDF combined with linear classifiers provides a computationally efficient and interpretable baseline for three-class answer grading. The approach scales well to large datasets and requires no GPU, making it accessible for rapid prototyping.

However, the bag-of-words assumption is a fundamental limitation for this task: the model ignores word order, syntax, and the semantic relationship between the answer and its originating question. Two answers with similar vocabulary but different relevance to the question would receive near-identical feature representations. Future work could address this by incorporating the question text as a second input, using sentence-pair models such as fine-tuned BERT or RoBERTa, which encode cross-sequence attention and have achieved state-of-the-art results on CQA benchmarks.

6. Conclusion

This project presented a multi-class text classification approach to automatic answer grading using the SemEval-2013 CQA dataset. We implemented a full NLP pipeline covering preprocessing, TF-IDF vectorization, and training of two linear classifiers (Logistic Regression and LinearSVC) to predict three answer quality levels: Bad (-1), Neutral (0), and Good (1).

Both models were evaluated using accuracy, precision, recall, weighted F1, and a 3×3 confusion matrix. Our best model (LinearSVC) achieved a weighted F1 of 0.634110 on the test set. Error analysis identified short answers, formally-written bad answers, and rare vocabulary as the primary sources of misclassification, motivating the use of contextual language models as a future direction.

References

- [1] Nakov, P., Marquez, L., Magdy, W., Moschitti, A., Glass, J., & Randeree, B. (2015). SemEval-2015 Task 3: Answer Selection in Community Question Answering. In Proceedings of SemEval-2015.
- [2] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- [3] Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.

[4] Joachims, T. (1998). Text Categorization with Support Vector Machines. ECML 1998.